for operations on arrays and lists, and the difference between singly- and doubly-linked lists.

So, imagine that you only have to keep an ordering on the cache entries, being able to go from the least to the most recently used.

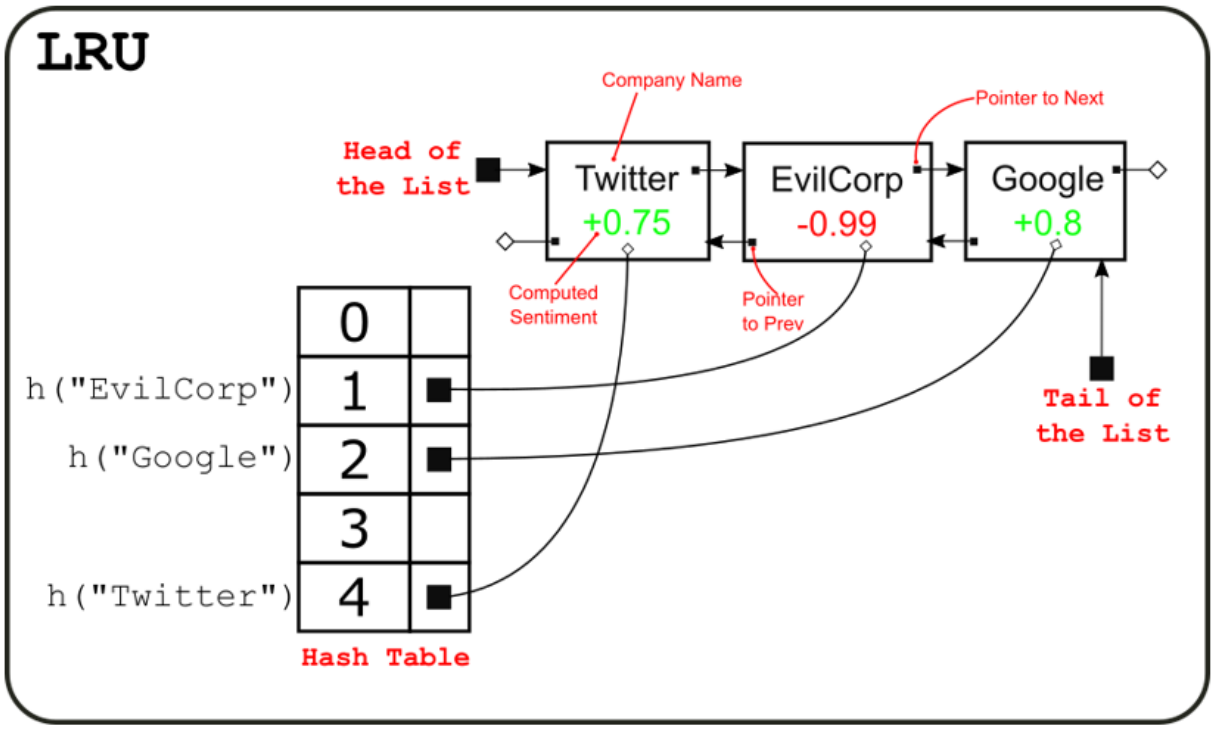


Figure 7.5 Structure of an LRU cache.
You can see the tree data elements that are stored for the cache and need to be updated after every operation:
1. The hash table;
2. The head of a doubly-linked list.
3. A pointer to the last element in the list.
Notice how each element in the hash table points to a node in the list, where all the data is stored. To get from a list entry to the corresponding hash entry, we have to hash the name of the company stored in the